TYPE(REFERENCE_MANUAL) :: CODATA = V2.5.3

! Fortran library for fundamental physical constants

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Part I

GETTING STARTED

Chapter 1

INTRODUCTION

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants published by the CODATA. A C API allows usage from C, or can be used as an interoperability layer for other languages. A Python wrapper allows easy usage from Python.

The constants are implemented as derived type which carries the name, the value, the uncertainty and the unit. All the *codata* constants are provided as double precision reals. The names are quite long and can be aliased with shorter names.

FEATURES

It provides:

- *codata* constants 2022
- *codata* constants 2018
- *codata* constants 2014
- *codata* constants 2010

DOCUMENTATION

POSIX-style man-pages are generated from the sources using the prep(1) preprocessor and are provided with the binaries. The API is written as man-pages and is basically a list of the constants.

MAN milanskocic.github.io.codata/man

PDF milanskocic.github.io/codata/man/codata-bookman.pdf

The man-pages are included in the reference manual.

PDF milanskocic.github.io/codata/latex/codata-refman.pdf

HTML milanskocic.github.io/codata/index.html

NOTES FOR FORTRAN USERS

The *codata* constant from 2022 were integrated in the Fortran standard library (stdlib). This library is complementary to the constants defined in the stdlib by providing older values for the constants. See github.com/fortran-lang/stdlib/releases/tag/v0.7.0. If you need older values for the CODATA constants you can use this library otherwise use the standard library.

To use *codata* within your fpm(1) (github.com/fortran-lang/fpm) project, add the following lines to your file:

```
[dependencies]
codata = { git="https://github.com/MilanSkocic/codata.git" }
```

If you only need sources for the *codata* constants that you can integrate directly embed in your sources you may be interested by this repository github.com/vmagnin/fundamental_constants.

LICENSE

MIT

Chapter 2

INSTALLATION

BINARIES

Download the binaries for your operating system from github.com/MilanSkocic/codata/releases. The archives contains 3 folders bin, lib and share that you can simply copy/paste somewhere on your PATH:

- Unix-like: $HOME/.local
- Windows: add .local to your %USERPROFILE% and add it to your PATH

SOURCES

Building the static and shared libraries requires the following dependencies:

- prep(1) >=9.2.0: 20220814
- gfortran(1) >=10
- gcc(1) >=10
- fpm(1) >=0.12
- python(1) >=3.10

```
$ . ./configure
$ make sources
$ make
$ make install [prefix=path]
```

Building the Python wrapper requires additional dependencies:

- python interpreter: 3.10, 3.11, 3.12, 3.13, 3.14

```
$ make python
```

Install the wheels in the py/sdist/ folder with pip.

Building the documentation requires even more additional dependencies:

- txt2man(1) >=1.7
- bookman(1) >=1.7
- txt2latex(1) >=0.3
- pdflatex(1) >=2025
- make4ht(1) >=0.4c
- sphinx >=8.0
- ford(1) >=7.0

Chapter 3

EXAMPLES

Minimal working examples in Fortran, C and Python are presented below. The use of the constants is pretty straightforward:

- import the module or include the header
- access the constant members: name, value, uncertainty, unit

FORTRAN

```fortran
program example_in_f
use codata
implicit none
print '(A)', '########## EXAMPLE IN FORTRAN ##########'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ",  SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ",  MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ",  MOLAR_MASS_CONSTANT_2010%value
end program
```

C

```c
#include <stdio.h>
#include "codata.h"
int main(void){
printf("########## EXAMPLE IN C ##########\n");
printf("%s\n","# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n","# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
```

```
printf("%s\n","# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s\n","# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}
```

PYTHON

```python
import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("########## EXAMPLE IN PYTHON ##########")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

Part II

API

Chapter 4

CODATA_CLI(1)

NAME

codata - fundamental physical constants

SYNOPSIS

`codata [OPTION...]  [REGEX_PATTERN...]`

DESCRIPTION

codata(1) is a command line interface which prints all the codata constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with REGEX PATTERNS.

OPTIONS

–year, -y YEAR Codata constants: 2022, 2018, 2014, 2010

–pattern, -p PATTERN Regex pattern for filtering the constants.

–value, -a Show only the value.

–error, -e Show only the uncertainty.

–usage Show usage text and exit.

–version, -v Show version information and exit.

–help, -h Show help text and exit.

NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see urbanjost.github.io/M_CLI2/set_args.3m_cli2.html

EXAMPLE

Minimal example

```
codata
codata -y 2018 molar electron
codata -y 2014 -p molar.*gas,electron.*eV
codata [B,b]oltzmann.*eV
```

SEE ALSO

codata(3)

Chapter 5

CODATA(3)

NAME

codata - libray for fundamental physical constants

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

`Fortran:  use codata`

`C: include "codata.h"`

`Python:  import pycodata`

DESCRIPTION

codata(3) is a Fortran library providing the fundamental physical constants published by the CODATA.

Fortran API exposes the constants as derived type codata_constant_type which carries:

- name
- value
- uncertainty
- unit

C API exposes a structure codata_constant_type that defines the same members as in Fortran:

- name
- value
- uncertainty
- unit

Python exposes a dictionary with the keys being:

- the name

- the value
- the uncertainty
- the unit

NOTES

The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix. All constants are declared with the same variable name in Fortran, C, and Python.

EXAMPLE

Example in Fortran

```
program example_in_f
use codata
implicit none
print '(A)', '########## EXAMPLE IN FORTRAN ##########'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ",  SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ",  MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ",  MOLAR_MASS_CONSTANT_2010%value
end program
```

Example in C

```
#include <stdio.h>
#include "codata.h"


int main(void){
printf("%f", c);
printf("########## EXAMPLE IN C ##########\n");
printf("%s\n","# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n","# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s\n","# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
```

```
printf("%s\n","# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}
```

Example in Python

```
import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("########## EXAMPLE IN PYTHON ##########")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

SEE ALSO

codata(1), codata_version(3), codata_constant_type(3)

CODATA 2022

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS

- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY
- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY
- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT
- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T

- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90
- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS

- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM
- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP
- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP
- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP

- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY
- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP
- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133

- INVERSE_FINE_STRUCTURE_CONSTANT
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP
- JOULE_ELECTRON_VOLT_RELATIONSHIP
- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP
- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS_EFFICACY
- MAG_FLUX_QUANTUM

- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
- MOLAR_VOLUME_OF_SILICON
- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO
- MUON_G_FACTOR
- MUON_MAG_MOM
- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
- MUON_TAU_MASS_RATIO
- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S
- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV
- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY
- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO

- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO
- NEUTRON_PROTON_MASS_DIFFERENCE
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C
- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME
- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH

- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO
- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
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- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO
- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS
- PROTON_TAU_MASS_RATIO
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- QUANTUM_OF_CIRCULATION_TIMES_2
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- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH
- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
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- RYDBERG_CONSTANT_TIMES_HC_IN_EV
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- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
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- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
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- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
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- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT
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- TAU_MOLAR_MASS
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- TRITON_PROTON_MASS_RATIO
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- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
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- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018
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- VACUUM_ELECTRIC_PERMITTIVITY_2018
- VACUUM_MAG_PERMEABILITY_2018
- VON_KLITZING_CONSTANT_2018
- WEAK_MIXING_ANGLE_2018
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2018
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2018
- W_TO_Z_MASS_RATIO_2018

CODATA 2014

List of available constants:

- LATTICE_SPACING_OF_SILICON_2014
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2014
- ALPHA_PARTICLE_MASS_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ALPHA_PARTICLE_MASS_IN_U_2014
- ALPHA_PARTICLE_MOLAR_MASS_2014
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2014
- ANGSTROM_STAR_2014
- ATOMIC_MASS_CONSTANT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2014
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2014
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2014
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2014
- ATOMIC_UNIT_OF_ACTION_2014
- ATOMIC_UNIT_OF_CHARGE_2014
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2014
- ATOMIC_UNIT_OF_CURRENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2014
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2014
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2014
- ATOMIC_UNIT_OF_ENERGY_2014
- ATOMIC_UNIT_OF_FORCE_2014
- ATOMIC_UNIT_OF_LENGTH_2014
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2014

- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2014
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2014
- ATOMIC_UNIT_OF_MASS_2014
- ATOMIC_UNIT_OF_MOMUM_2014
- ATOMIC_UNIT_OF_PERMITTIVITY_2014
- ATOMIC_UNIT_OF_TIME_2014
- ATOMIC_UNIT_OF_VELOCITY_2014
- AVOGADRO_CONSTANT_2014
- BOHR_MAGNETON_2014
- BOHR_MAGNETON_IN_EV_T_2014
- BOHR_MAGNETON_IN_HZ_T_2014
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- BOHR_MAGNETON_IN_K_T_2014
- BOHR_RADIUS_2014
- BOLTZMANN_CONSTANT_2014
- BOLTZMANN_CONSTANT_IN_EV_K_2014
- BOLTZMANN_CONSTANT_IN_HZ_K_2014
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2014
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2014
- CLASSICAL_ELECTRON_RADIUS_2014
- COMPTON_WAVELENGTH_2014
- COMPTON_WAVELENGTH_OVER_2_PI_2014
- CONDUCTANCE_QUANTUM_2014
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2014
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2014
- CU_X_UNIT_2014
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2014
- DEUTERON_ELECTRON_MASS_RATIO_2014
- DEUTERON_G_FACTOR_2014
- DEUTERON_MAG_MOM_2014
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- DEUTERON_MASS_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- DEUTERON_MASS_IN_U_2014
- DEUTERON_MOLAR_MASS_2014

- DEUTERON_NEUTRON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MASS_RATIO_2014
- DEUTERON_RMS_CHARGE_RADIUS_2014
- ELECTRIC_CONSTANT_2014
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2014
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2014
- ELECTRON_DEUTERON_MASS_RATIO_2014
- ELECTRON_G_FACTOR_2014
- ELECTRON_GYROMAG_RATIO_2014
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2014
- ELECTRON_HELION_MASS_RATIO_2014
- ELECTRON_MAG_MOM_2014
- ELECTRON_MAG_MOM_ANOMALY_2014
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- ELECTRON_MASS_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ELECTRON_MASS_IN_U_2014
- ELECTRON_MOLAR_MASS_2014
- ELECTRON_MUON_MAG_MOM_RATIO_2014
- ELECTRON_MUON_MASS_RATIO_2014
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2014
- ELECTRON_NEUTRON_MASS_RATIO_2014
- ELECTRON_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_PROTON_MASS_RATIO_2014
- ELECTRON_TAU_MASS_RATIO_2014
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2014
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2014
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_TRITON_MASS_RATIO_2014
- ELECTRON_VOLT_2014
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2014
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2014
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2014

- ELECTRON_VOLT_JOULE_RELATIONSHIP_2014
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2014
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2014
- ELEMENTARY_CHARGE_2014
- ELEMENTARY_CHARGE_OVER_H_2014
- FARADAY_CONSTANT_2014
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2014
- FERMI_COUPLING_CONSTANT_2014
- FINE_STRUCTURE_CONSTANT_2014
- FIRST_RADIATION_CONSTANT_2014
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2014
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2014
- HARTREE_ENERGY_2014
- HARTREE_ENERGY_IN_EV_2014
- HARTREE_HERTZ_RELATIONSHIP_2014
- HARTREE_INVERSE_METER_RELATIONSHIP_2014
- HARTREE_JOULE_RELATIONSHIP_2014
- HARTREE_KELVIN_RELATIONSHIP_2014
- HARTREE_KILOGRAM_RELATIONSHIP_2014
- HELION_ELECTRON_MASS_RATIO_2014
- HELION_G_FACTOR_2014
- HELION_MAG_MOM_2014
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- HELION_MASS_2014
- HELION_MASS_ENERGY_EQUIVALENT_2014
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- HELION_MASS_IN_U_2014
- HELION_MOLAR_MASS_2014
- HELION_PROTON_MASS_RATIO_2014
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2014
- HERTZ_HARTREE_RELATIONSHIP_2014
- HERTZ_INVERSE_METER_RELATIONSHIP_2014
- HERTZ_JOULE_RELATIONSHIP_2014
- HERTZ_KELVIN_RELATIONSHIP_2014

- HERTZ_KILOGRAM_RELATIONSHIP_2014
- INVERSE_FINE_STRUCTURE_CONSTANT_2014
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2014
- INVERSE_METER_HARTREE_RELATIONSHIP_2014
- INVERSE_METER_HERTZ_RELATIONSHIP_2014
- INVERSE_METER_JOULE_RELATIONSHIP_2014
- INVERSE_METER_KELVIN_RELATIONSHIP_2014
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2014
- INVERSE_OF_CONDUCTANCE_QUANTUM_2014
- JOSEPHSON_CONSTANT_2014
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2014
- JOULE_HARTREE_RELATIONSHIP_2014
- JOULE_HERTZ_RELATIONSHIP_2014
- JOULE_INVERSE_METER_RELATIONSHIP_2014
- JOULE_KELVIN_RELATIONSHIP_2014
- JOULE_KILOGRAM_RELATIONSHIP_2014
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2014
- KELVIN_HARTREE_RELATIONSHIP_2014
- KELVIN_HERTZ_RELATIONSHIP_2014
- KELVIN_INVERSE_METER_RELATIONSHIP_2014
- KELVIN_JOULE_RELATIONSHIP_2014
- KELVIN_KILOGRAM_RELATIONSHIP_2014
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2014
- KILOGRAM_HARTREE_RELATIONSHIP_2014
- KILOGRAM_HERTZ_RELATIONSHIP_2014
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2014
- KILOGRAM_JOULE_RELATIONSHIP_2014
- KILOGRAM_KELVIN_RELATIONSHIP_2014
- LATTICE_PARAMETER_OF_SILICON_2014
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2014
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2014
- MAG_CONSTANT_2014
- MAG_FLUX_QUANTUM_2014

- MOLAR_GAS_CONSTANT_2014
- MOLAR_MASS_CONSTANT_2014
- MOLAR_MASS_OF_CARBON_12_2014
- MOLAR_PLANCK_CONSTANT_2014
- MOLAR_PLANCK_CONSTANT_TIMES_C_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2014
- MOLAR_VOLUME_OF_SILICON_2014
- MO_X_UNIT_2014
- MUON_COMPTON_WAVELENGTH_2014
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- MUON_ELECTRON_MASS_RATIO_2014
- MUON_G_FACTOR_2014
- MUON_MAG_MOM_2014
- MUON_MAG_MOM_ANOMALY_2014
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- MUON_MASS_2014
- MUON_MASS_ENERGY_EQUIVALENT_2014
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- MUON_MASS_IN_U_2014
- MUON_MOLAR_MASS_2014
- MUON_NEUTRON_MASS_RATIO_2014
- MUON_PROTON_MAG_MOM_RATIO_2014
- MUON_PROTON_MASS_RATIO_2014
- MUON_TAU_MASS_RATIO_2014
- NATURAL_UNIT_OF_ACTION_2014
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2014
- NATURAL_UNIT_OF_ENERGY_2014
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2014
- NATURAL_UNIT_OF_LENGTH_2014
- NATURAL_UNIT_OF_MASS_2014
- NATURAL_UNIT_OF_MOMUM_2014
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2014
- NATURAL_UNIT_OF_TIME_2014
- NATURAL_UNIT_OF_VELOCITY_2014
- NEUTRON_COMPTON_WAVELENGTH_2014

- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2014
- NEUTRON_ELECTRON_MASS_RATIO_2014
- NEUTRON_G_FACTOR_2014
- NEUTRON_GYROMAG_RATIO_2014
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2014
- NEUTRON_MAG_MOM_2014
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- NEUTRON_MASS_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_MASS_IN_U_2014
- NEUTRON_MOLAR_MASS_2014
- NEUTRON_MUON_MASS_RATIO_2014
- NEUTRON_PROTON_MAG_MOM_RATIO_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2014
- NEUTRON_PROTON_MASS_RATIO_2014
- NEUTRON_TAU_MASS_RATIO_2014
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2014
- NUCLEAR_MAGNETON_2014
- NUCLEAR_MAGNETON_IN_EV_T_2014
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- NUCLEAR_MAGNETON_IN_K_T_2014
- NUCLEAR_MAGNETON_IN_MHZ_T_2014
- PLANCK_CONSTANT_2014
- PLANCK_CONSTANT_IN_EV_S_2014
- PLANCK_CONSTANT_OVER_2_PI_2014
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2014
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2014
- PLANCK_LENGTH_2014
- PLANCK_MASS_2014

- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2014
- PLANCK_TEMPERATURE_2014
- PLANCK_TIME_2014
- PROTON_CHARGE_TO_MASS_QUOTIENT_2014
- PROTON_COMPTON_WAVELENGTH_2014
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- PROTON_ELECTRON_MASS_RATIO_2014
- PROTON_G_FACTOR_2014
- PROTON_GYROMAG_RATIO_2014
- PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- PROTON_MAG_MOM_2014
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- PROTON_MAG_SHIELDING_CORRECTION_2014
- PROTON_MASS_2014
- PROTON_MASS_ENERGY_EQUIVALENT_2014
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- PROTON_MASS_IN_U_2014
- PROTON_MOLAR_MASS_2014
- PROTON_MUON_MASS_RATIO_2014
- PROTON_NEUTRON_MAG_MOM_RATIO_2014
- PROTON_NEUTRON_MASS_RATIO_2014
- PROTON_RMS_CHARGE_RADIUS_2014
- PROTON_TAU_MASS_RATIO_2014
- QUANTUM_OF_CIRCULATION_2014
- QUANTUM_OF_CIRCULATION_TIMES_2_2014
- RYDBERG_CONSTANT_2014
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2014
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2014
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2014
- SECOND_RADIATION_CONSTANT_2014
- SHIELDED_HELION_GYROMAG_RATIO_2014
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_HELION_MAG_MOM_2014
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014

- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_PROTON_MAG_MOM_2014
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SPEED_OF_LIGHT_IN_VACUUM_2014
- STANDARD_ACCELERATION_OF_GRAVITY_2014
- STANDARD_ATMOSPHERE_2014
- STANDARD_STATE_PRESSURE_2014
- STEFAN_BOLTZMANN_CONSTANT_2014
- TAU_COMPTON_WAVELENGTH_2014
- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2014
- TAU_ELECTRON_MASS_RATIO_2014
- TAU_MASS_2014
- TAU_MASS_ENERGY_EQUIVALENT_2014
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TAU_MASS_IN_U_2014
- TAU_MOLAR_MASS_2014
- TAU_MUON_MASS_RATIO_2014
- TAU_NEUTRON_MASS_RATIO_2014
- TAU_PROTON_MASS_RATIO_2014
- THOMSON_CROSS_SECTION_2014
- TRITON_ELECTRON_MASS_RATIO_2014
- TRITON_G_FACTOR_2014
- TRITON_MAG_MOM_2014
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- TRITON_MASS_2014
- TRITON_MASS_ENERGY_EQUIVALENT_2014
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TRITON_MASS_IN_U_2014
- TRITON_MOLAR_MASS_2014
- TRITON_PROTON_MASS_RATIO_2014
- UNIFIED_ATOMIC_MASS_UNIT_2014

- VON_KLITZING_CONSTANT_2014
- WEAK_MIXING_ANGLE_2014
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2014
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2014

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List of available constants:

- LATTICE_SPACING_OF_SILICON_2010
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2010
- ALPHA_PARTICLE_MASS_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ALPHA_PARTICLE_MASS_IN_U_2010
- ALPHA_PARTICLE_MOLAR_MASS_2010
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2010
- ANGSTROM_STAR_2010
- ATOMIC_MASS_CONSTANT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2010
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2010
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_ACTION_2010
- ATOMIC_UNIT_OF_CHARGE_2010
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2010
- ATOMIC_UNIT_OF_CURRENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2010
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2010

- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2010
- ATOMIC_UNIT_OF_ENERGY_2010
- ATOMIC_UNIT_OF_FORCE_2010
- ATOMIC_UNIT_OF_LENGTH_2010
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2010
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2010
- ATOMIC_UNIT_OF_MASS_2010
- ATOMIC_UNIT_OF_MOMUM_2010
- ATOMIC_UNIT_OF_PERMITTIVITY_2010
- ATOMIC_UNIT_OF_TIME_2010
- ATOMIC_UNIT_OF_VELOCITY_2010
- AVOGADRO_CONSTANT_2010
- BOHR_MAGNETON_2010
- BOHR_MAGNETON_IN_EV_T_2010
- BOHR_MAGNETON_IN_HZ_T_2010
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- BOHR_MAGNETON_IN_K_T_2010
- BOHR_RADIUS_2010
- BOLTZMANN_CONSTANT_2010
- BOLTZMANN_CONSTANT_IN_EV_K_2010
- BOLTZMANN_CONSTANT_IN_HZ_K_2010
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2010
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2010
- CLASSICAL_ELECTRON_RADIUS_2010
- COMPTON_WAVELENGTH_2010
- COMPTON_WAVELENGTH_OVER_2_PI_2010
- CONDUCTANCE_QUANTUM_2010
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2010
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2010
- CU_X_UNIT_2010
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2010
- DEUTERON_ELECTRON_MASS_RATIO_2010
- DEUTERON_G_FACTOR_2010
- DEUTERON_MAG_MOM_2010
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010

- DEUTERON_MASS_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- DEUTERON_MASS_IN_U_2010
- DEUTERON_MOLAR_MASS_2010
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MASS_RATIO_2010
- DEUTERON_RMS_CHARGE_RADIUS_2010
- ELECTRIC_CONSTANT_2010
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2010
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2010
- ELECTRON_DEUTERON_MASS_RATIO_2010
- ELECTRON_G_FACTOR_2010
- ELECTRON_GYROMAG_RATIO_2010
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2010
- ELECTRON_HELION_MASS_RATIO_2010
- ELECTRON_MAG_MOM_2010
- ELECTRON_MAG_MOM_ANOMALY_2010
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- ELECTRON_MASS_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ELECTRON_MASS_IN_U_2010
- ELECTRON_MOLAR_MASS_2010
- ELECTRON_MUON_MAG_MOM_RATIO_2010
- ELECTRON_MUON_MASS_RATIO_2010
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2010
- ELECTRON_NEUTRON_MASS_RATIO_2010
- ELECTRON_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_PROTON_MASS_RATIO_2010
- ELECTRON_TAU_MASS_RATIO_2010
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2010
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2010
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_TRITON_MASS_RATIO_2010

- ELECTRON_VOLT_2010
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2010
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2010
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2010
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010
- ELEMENTARY_CHARGE_2010
- ELEMENTARY_CHARGE_OVER_H_2010
- FARADAY_CONSTANT_2010
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010
- FERMI_COUPLING_CONSTANT_2010
- FINE_STRUCTURE_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2010
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2010
- HARTREE_ENERGY_2010
- HARTREE_ENERGY_IN_EV_2010
- HARTREE_HERTZ_RELATIONSHIP_2010
- HARTREE_INVERSE_METER_RELATIONSHIP_2010
- HARTREE_JOULE_RELATIONSHIP_2010
- HARTREE_KELVIN_RELATIONSHIP_2010
- HARTREE_KILOGRAM_RELATIONSHIP_2010
- HELION_ELECTRON_MASS_RATIO_2010
- HELION_G_FACTOR_2010
- HELION_MAG_MOM_2010
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- HELION_MASS_2010
- HELION_MASS_ENERGY_EQUIVALENT_2010
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- HELION_MASS_IN_U_2010
- HELION_MOLAR_MASS_2010
- HELION_PROTON_MASS_RATIO_2010
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2010

- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2010
- HERTZ_HARTREE_RELATIONSHIP_2010
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- HERTZ_JOULE_RELATIONSHIP_2010
- HERTZ_KELVIN_RELATIONSHIP_2010
- HERTZ_KILOGRAM_RELATIONSHIP_2010
- INVERSE_FINE_STRUCTURE_CONSTANT_2010
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2010
- INVERSE_METER_HARTREE_RELATIONSHIP_2010
- INVERSE_METER_HERTZ_RELATIONSHIP_2010
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- INVERSE_METER_KELVIN_RELATIONSHIP_2010
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2010
- INVERSE_OF_CONDUCTANCE_QUANTUM_2010
- JOSEPHSON_CONSTANT_2010
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
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- JOULE_HERTZ_RELATIONSHIP_2010
- JOULE_INVERSE_METER_RELATIONSHIP_2010
- JOULE_KELVIN_RELATIONSHIP_2010
- JOULE_KILOGRAM_RELATIONSHIP_2010
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2010
- KELVIN_HARTREE_RELATIONSHIP_2010
- KELVIN_HERTZ_RELATIONSHIP_2010
- KELVIN_INVERSE_METER_RELATIONSHIP_2010
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- KELVIN_KILOGRAM_RELATIONSHIP_2010
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
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- KILOGRAM_INVERSE_METER_RELATIONSHIP_2010
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- KILOGRAM_KELVIN_RELATIONSHIP_2010

- LATTICE_PARAMETER_OF_SILICON_2010
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010
- MAG_FLUX_QUANTUM_2010
- MOLAR_GAS_CONSTANT_2010
- MOLAR_MASS_CONSTANT_2010
- MOLAR_MASS_OF_CARBON_12_2010
- MOLAR_PLANCK_CONSTANT_2010
- MOLAR_PLANCK_CONSTANT_TIMES_C_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2010
- MOLAR_VOLUME_OF_SILICON_2010
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- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2010
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- MUON_MAG_MOM_2010
- MUON_MAG_MOM_ANOMALY_2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
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- MUON_MASS_IN_U_2010
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- MUON_NEUTRON_MASS_RATIO_2010
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- MUON_PROTON_MASS_RATIO_2010
- MUON_TAU_MASS_RATIO_2010
- NATURAL_UNIT_OF_ACTION_2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2010
- NATURAL_UNIT_OF_ENERGY_2010
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- NEUTRON_ELECTRON_MASS_RATIO_2010
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- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010
- NEUTRON_PROTON_MASS_RATIO_2010
- NEUTRON_TAU_MASS_RATIO_2010
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2010
- NUCLEAR_MAGNETON_2010
- NUCLEAR_MAGNETON_IN_EV_T_2010
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- NUCLEAR_MAGNETON_IN_K_T_2010
- NUCLEAR_MAGNETON_IN_MHZ_T_2010
- PLANCK_CONSTANT_2010
- PLANCK_CONSTANT_IN_EV_S_2010

- PLANCK_CONSTANT_OVER_2_PI_2010
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2010
- PLANCK_LENGTH_2010
- PLANCK_MASS_2010
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2010
- PLANCK_TEMPERATURE_2010
- PLANCK_TIME_2010
- PROTON_CHARGE_TO_MASS_QUOTIENT_2010
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- PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- PROTON_ELECTRON_MASS_RATIO_2010
- PROTON_G_FACTOR_2010
- PROTON_GYROMAG_RATIO_2010
- PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- PROTON_MAG_MOM_2010
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- PROTON_MAG_SHIELDING_CORRECTION_2010
- PROTON_MASS_2010
- PROTON_MASS_ENERGY_EQUIVALENT_2010
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- PROTON_MASS_IN_U_2010
- PROTON_MOLAR_MASS_2010
- PROTON_MUON_MASS_RATIO_2010
- PROTON_NEUTRON_MAG_MOM_RATIO_2010
- PROTON_NEUTRON_MASS_RATIO_2010
- PROTON_RMS_CHARGE_RADIUS_2010
- PROTON_TAU_MASS_RATIO_2010
- QUANTUM_OF_CIRCULATION_2010
- QUANTUM_OF_CIRCULATION_TIMES_2_2010
- RYDBERG_CONSTANT_2010
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2010
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2010
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2010

- SECOND_RADIATION_CONSTANT_2010
- SHIELDED_HELION_GYROMAG_RATIO_2010
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_HELION_MAG_MOM_2010
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_PROTON_MAG_MOM_2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SPEED_OF_LIGHT_IN_VACUUM_2010
- STANDARD_ACCELERATION_OF_GRAVITY_2010
- STANDARD_ATMOSPHERE_2010
- STANDARD_STATE_PRESSURE_2010
- STEFAN_BOLTZMANN_CONSTANT_2010
- TAU_COMPTON_WAVELENGTH_2010
- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2010
- TAU_ELECTRON_MASS_RATIO_2010
- TAU_MASS_2010
- TAU_MASS_ENERGY_EQUIVALENT_2010
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TAU_MASS_IN_U_2010
- TAU_MOLAR_MASS_2010
- TAU_MUON_MASS_RATIO_2010
- TAU_NEUTRON_MASS_RATIO_2010
- TAU_PROTON_MASS_RATIO_2010
- THOMSON_CROSS_SECTION_2010
- TRITON_ELECTRON_MASS_RATIO_2010
- TRITON_G_FACTOR_2010
- TRITON_MAG_MOM_2010
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- TRITON_MASS_2010
- TRITON_MASS_ENERGY_EQUIVALENT_2010

- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TRITON_MASS_IN_U_2010
- TRITON_MOLAR_MASS_2010
- TRITON_PROTON_MASS_RATIO_2010
- UNIFIED_ATOMIC_MASS_UNIT_2010
- VON_KLITZING_CONSTANT_2010
- WEAK_MIXING_ANGLE_2010
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2010
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

Chapter 6

CODATA_VERSION(3)

NAME

version - get the version of the Fortran library codata

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

`Fortran: character(len=:), pointer version()`

`C: char* codata_version()`

`Python: pycodata.__version__`

DESCRIPTION

Get the library version.

EXAMPLE

Fortran:

```
use codata
print *, "version = ", version()
```

C:

```
include <stdio.h>
include "codata.h"
printf("version = %s\n", codata_version());
```

Python:

```
import pycodata
print(f"version = {pycodata.__version__}")
```

SEE ALSO

codata(3)

Chapter 7

CODATA_CONSTANT_TYPE(3)

NAME

codata_constant_type - data type defining a constant

SYNOPSIS

`Fortran: type(codata_constant_type) :: c`

`C: struct codata_constant_type c`

DESCRIPTION

Defines a constant with the name, the value, the uncertainty and the unit.

EXAMPLE

Fortran:

```
use codata
type(codata_constant_type), pointer :: c
c = SPEED_OF_LIGHT_IN_VACUUM
```

C:

```
struct codata_constant_type c;
c = SPEED_OF_LIGHT_IN_VACUUM;
```

SEE ALSO

codata(3)

Part III

APPENDICES

Chapter 8

CHANGELOG

VERSION 2.5.3

- Maintenance release. NO API break.
- Complete rewrite of the documentation structure.
- The API is now written as man-pages:
  - codata(1)
  - codata(3)
  - codata_version(3)
  - codata_constant_type(3)

VERSION 2.5.2

- Maintenance release. No API break.
- Use prep(1) for auto-generating the version parameter.
- Add new API for getting the version number:
  - version() instead of get_version().
  - get_version() is marked as deprecated, it will be removed in the next major release 3.0 when the new codata constants will be released i.e. 2026/2027.

VERSION 2.5.1

- Code refactoring and cleaning: trailing spaces, empty lines, formatted comments.
- Remove specific modules for C API: defined in the same module as Fortran equivalents.
- No API break.

VERSION 2.5.0

- Add a response_file (urbanjost.github.io/M_CLI2/set_args.3m_cli2.html) capability for the CLI.
- Add rst output for the man pages.
- Integration of the man pages in the online documentation generated with sphinx.

VERSION 2.4.1

- Maintenance release.
- Fix option formatting in help text and man page.
- Refactoring

VERSION 2.4.0

- Add tables for each codata year containing all constants.
- Add CLI codata(1) for searching constants.
- Refactoring code.

VERSION 2.3.1

- Refactoring the configure.sh script.
- Remove support for 3.14t. No official release on python.org.
- If binaries for Python 3.14t are needed you need to compile them by yourself.

VERSION 2.3.0

- Remove support for Python 3.9 and add support for Python 3.14(t).

VERSION 2.2.0

- Switch to UCRT64 for Windows binaries.
- Switch to sphinx documentation using fspxh (ttps://github.com/jalvesz/fspx).
- Update references with publication for codata 2022.
- Update compilation flags for compatibility with stdlib.

VERSION 2.1.1

- No code change.
- Code refractoring and cleaning
- Update CI/CD workflows.

VERSION 2.1.0

- Roll back to C API in Fortran code: easier maintenance.
- Roll back to compiled C extension for python: easier maintenance.

VERSION 2.0.1

- Fix bug in version for Fortran code.

VERSION 2.0.0

- Drop compiled extensions for Python.
- Pure Python code for constants auto-generated as it is the case for the Fortran code.
- Pure C code for constants auto-generated as it is the case for the Fortran code.
- API break:
  - No more C API in the Fortran code.
  - Use the pure C code to build a C library.

VERSION 1.2.2

- Fix conflict that could occur with C API modules. Add prefix in module names.
- Cleanup and refractoring.
- Documentation update.

VERSION 1.2.1

- Refractoring
- Merge back C API and python wrapper.

VERSION 1.2.0

- Refractoring
- Documentation update.

VERSION 1.1.0

- C API and Python wrapper moved to their own repositories.
- API break: C API is no more provided by default. Use the optional C wrapper.
- Code cleanup
- Documentation update

VERSION 1.0.0

- Add codata values for 2010, 2014 and 2018.
- Code refractoring and code cleaning.
- Documentation update and switch to only FORD documentation.
- Rewrite code generators in python.
- Generate source code for stdlib.
- API break: constants are defined as DT like in stdlib.

VERSION 0.10.0

- Remove remove generation of the version module.

- Add tests using the test-drive framework.
- Explicit parameter constants for Fortran and protected constants for C API.
- Minor fixes in documentation.
- Code cleanup.
- Merge of all code for autogeneration in one file.

VERSION 0.9.0

- No API changes.
- Automatic generation of the version module.
- Generic Makefiles for automatic the building process of the library and the pywrapper.
- Add targets: build, build_debug, test, test_debug.
- Minor fixes in documentation.

VERSION 0.8.2

- No API changes.
- Improve Makefile for generating the source code at each compilation.
- Source generator rewritten in Fortran.
- Switch to pyproject.toml for the Python wrapper.
- Minor fixes in documentation.

VERSION 0.8.1

- Use shared library in python wrapper.
- Minor fixes in documentation.

VERSION 0.8.0

- Back to the approach with a library.
- Compatible with fpm.
- Configuration file for setting all the environmental variables.
- Global makefile for building a static library (through fpm) and a shared library.
- Automatic copy of the necessary sources for the python wrapper.
- Python wrapper built with the static library
  - no dependency on a shared library.
  - sources and static library embeded in the python wrapper.
- FORD for documenting the Fortran code.
- Integration of the FORD documentation into the main documentation with sphinx.

VERSION 0.7.1

- Minor fixes in generator code
- Add automatic copy of c sources for the python wrapper.

VERSION 0.7.0

- Migrate documentation from doxygen to sphinx+breathe.
- Add YEAR constant indicating the year of the codata constants.
- Refractoring

VERSION 0.6.0

- Created documentation.
- Fixed missing uncertainties for Cpython.

VERSION 0.5.0

- Changed the complete approach by not generating a library but only source files for different languages.
- Available languages: Fortran, C, python, CPython

VERSION 0.4.0

- Bring back pywrapper in the codata repository to sync versions.
- Improvements of the documentation.

VERSION 0.3.0

- Only last codata constants.

VERSION 0.2.1

- Integration of Intel Fortran compiler and MSVC in cmake scripts.
- Add specifications and instructions for compiling on Windows

VERSION 0.2.0

- Bug fixes for the codata 2010.
- Bug fixes in the tests linked to the codata 2010.
- Add python wrapper for the number of constants method.

VERSION 0.1.0

Implementation of:

- the parser of the codata raw data

- the generator of the Fortran modules
- the C API and C header
- the python wrapper (will be moved to its repository next release).

Chapter 9

LICENSE

Bibliography

[1] P. J. Mohr, B. N. Taylor, and D. B. Newell, "CODATA recommended values of the fundamental physical constants: 2010," *Review of Modern Physics*, vol. 84, 2012.

[2] P. J. Mohr, B. N. Taylor, and D. B. Newell, "CODATA recommended values of the fundamental physical constants: 2014," *Journal of Physical and Chemical Reference Data*, vol. 45, 2016.

[3] P. J. Mohr, B. N. Taylor, and D. B. Newell, "CODATA recommended values of the fundamental physical constants: 2018," *Review of Modern Physics*, vol. 93, 2021.

[4] P. Mohr, D. Newell, B. Taylor, and E. Tiesinga. "CODATA Recommended Values of the Fundamental Physical Constants: 2022." version 1, Accessed: May 5, 2025. [Online]. Available: `https://arxiv.org/abs/2409.03787`, pre-published.

[5] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, "CODATA recommended values of the fundamental physical constants: 2022," *Reviews of Modern Physics*, vol. 97, no. 2, p. 025 002, 2025.